Graph-Structured Memory for Cognitive Agents: From Knowledge Graphs to Non-Euclidean Geometry

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Abstract

Large language model based agents require persistent memory that spans sessions, tracks temporal evolution, and supports multi-hop reasoning. As context windows grow, the naïve strategy is to compress all prior interaction into the prompt, however compression discards relational structure. Graph-structured memory offers a fundamentally different approach: store knowledge in a graph and retrieve selectively. Yet current systems embed these graphs exclusively in Euclidean space, incurring distortion on the hierarchical, tree-like structures inherent to agent memory. We survey research papers between 2023 and 2026, identifying six architectural patterns and a retrieval spectrum from training-free graph algorithms to graph neural network message passing. We then analyze how non-Euclidean geometric techniques, such as discrete Ricci curvature for diagnosing retrieval bottlenecks, hyperbolic embeddings for hierarchy-preserving retrieval, Lie group manifolds for temporal knowledge graph fusion, and mixed-curvature product spaces, are already proven in retrieval and KG reasoning, with significant gains over Euclidean baselines. We argue that since agent memory is fundamentally retrieval over a persistent knowledge graph, these techniques are directly applicable as foundational layers, but integrating them into cross-session memory architectures is the defining open challenge.

Keywords

Knowledge Graphs, Agent Memory, Non-Euclidean Geometry, Hyperbolic Embeddings, Temporal Knowledge Graphs, LLM Agents

1. Introduction

The dominant memory pattern for Large Language Model (LLM) agents is stateless: each session begins from scratch. When persistence is added, it typically takes one of two forms: i) flat extraction, i.e. triplet stores, key-value memories, or embedding-indexed chunks [2, 3]; or ii) brute-force compression of all prior interaction into the context window, which grows ever larger but still cannot preserve relational structure. Both handle factual recall but fail systematically on temporal reasoning, multi-hop inference, and knowledge evolution. Flat paradigms, such as vector stores, key-value extraction, and even the emerging LLM Wiki pattern [53], lack native support for multi-hop traversal, temporal ordering, causal structure, and hierarchical consolidation; graph-based architectures provide all of these by construction. Benchmark evidence quantifies the deficit: LoCoMo-Plus [4] reports cognitive accuracy below 25% across Mem0, Letta, and Zep, with a 35–45 pp drop from factual to cognitive tasks. MemoryAgentBench [5] finds all methods below 7% on multi-hop conflict resolution, while multi-strategy systems combining graph traversal with vector search achieve 91.4% retrieval accuracy versus 49.0% for extraction-based systems [6]. Graph-structured memory offers a fundamentally different strategy: instead of compressing everything into the context window, store knowledge in a graph and retrieve selectively. Knowledge graphs (KG) ground entities and relations while temporal KGs add validity intervals; hierarchical event graphs capture episodic consolidation. The core challenge then shifts from *what to discard* to *how to find the right subgraph*, i.e. a retrieval problem where graph topology is a first-class signal. We address the following research question: *How do current graph-based memory architectures handle retrieval, temporal reasoning, and structural representation, and can non-Euclidean geometric techniques address their identified limitations?* We survey the current state of the art in graph-based agent memory

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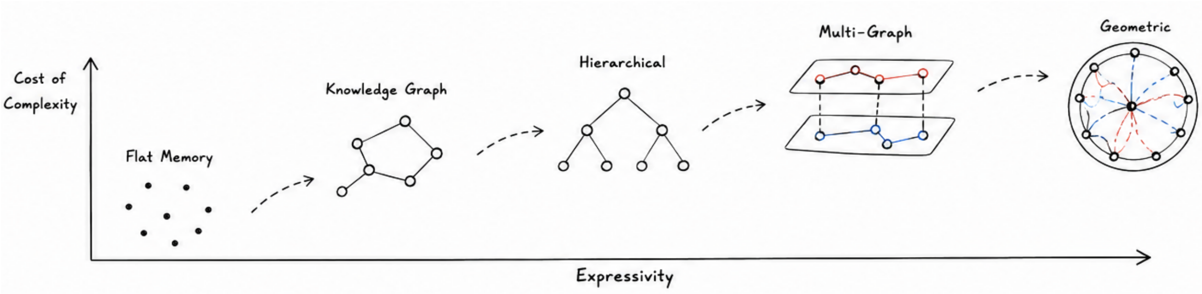

Figure 1: Structural progression of agent memory representations. Each level adds expressivity at the cost of complexity. Current production systems operate at the knowledge graph and hierarchical levels, while geometric representations remain a largely unexplored frontier.

systems between 2023 and 2026 [7, 8]. To scope this survey, we searched Semantic Scholar, arXiv, and the proceedings of NeurIPS, ICLR, ACL, EMNLP, KDD, and IJCAI (2023–2026) using the query terms *agent memory*, *graph memory*, *knowledge graph + LLM agent*, and *persistent memory + LLM*. We included papers that propose or evaluate a graph-structured memory component for LLM-based agents and excluded systems that use graphs solely for retrieval-augmented generation without a persistent memory loop. This yielded 30+ primary systems, which we organize by architectural pattern, temporal mechanism, and retrieval strategy. Three significant findings stand out: i) entity-relation KGs account for 60% of surveyed graph-memory papers; non-KG structures remain rare; ii) 57% implement no temporal mechanism, constituting a field-wide deficit; and iii) retrieval is overwhelmingly training-free, i.e. graph algorithms and multi-strategy heuristics power every production system, while only a tiny fraction of surveyed research work applies graph neural networks (GNN) message passing.

Figure 1 illustrates the structural progression from flat memory to geometric representations. This paper makes two types of contributions. Our survey, therefore, first provides: i) a taxonomy of six graph-based agent memory patterns including temporal grounding (Section 2); and ii) a retrieval spectrum from training-free graph algorithms to learned GNN retrieval, with a model-size equalizer analysis (Section 3). On the basis of this survey we derive: i) an argument, supported by converging evidence from geometric deep learning, that non-Euclidean techniques are directly applicable to the structural limitations identified in the survey (Section 4); and ii) a research agenda for geometry-aware persistent agent memory (Section 5).

2. Taxonomy of Graph-Based Agent Memory

From our classification we identify four established architectural patterns, two emerging ones, and a cross-cutting temporal dimension (Table 1). The most prevalent pattern is the *entity-relation knowledge graph*, representing memory as typed $(subject, relation, object)$ triples. CraniMem [13] combines a bounded FIFO episodic buffer with an entity-relation KG. UltRAG [14] queries large-scale KGs via a pre-trained GNN foundation model. AriGraph [15] integrates episodic and semantic memory within a single KG. GraphRAG [12] clusters via Leiden into hierarchical community summaries. Entity-relation KGs excel at factual memory but are limited to binary relations.

The *Hierarchical event graphs* pattern addresses this by maintaining episodic and semantic layers with cross-layer edges. GAM [16] combines a Topic Associative Network with Event Progression Graphs. MemFly [17] implements a three-layer hierarchy via the Information Bottleneck principle. [18] introduces Cue-Tag-Content heterogeneous graphs. MemGraphRAG [1] organizes memory into three layers (ontology schemas, entity-relation facts, source passages) with multi-agent self-adjudication, i.e. a conflict detection agent identifies inconsistencies and a resolution agent adjudicates via provenance evidence, demonstrating that filtering low-frequency triples actually improves accuracy on multi-hop Question Answering.

Bipartite and associative graphs take a lighter approach. For example, LP-RAG [19] constructs a

Table 1
Comparison of graph-based agent memory systems. Temporal: ✓ = deep mechanism, △ = timestamps/decay, — = none. Retrieval: GA = graph algorithms, MS = multi-strategy, RL = reinforcement learning, GNN = message passing, AB = agent-based.

System	Graph Type	Temp.	Retr.	Evolution	Best Score	Benchmark
Production & multi-strategy systems						
Zep/Graphiti	KG (bi-temporal)	✓	MS	Temporal invalidation	94.8%	Deep Mem. Retrieval
Hindsight	KG + vector	△	MS	—	91.4%	LongMemEval
FalkorDB+Mem0	KG (graph DB)	—	MS	Dedup/merge	<140 ms p99	Production
Hierarchical & multi-graph architectures						
GAM	Hierarchical (2-layer)	△	GA	State-based consol.	40.00 F1	LoCoMo (7B)
MAGMA	Multi-graph (4 types)	✓	GA	Dual-stream consol.	0.700	LoCoMo Judge
MemFly	Hierarchical (3-layer)	—	GA	IB clustering	44.4 F1	LoCoMo (GPT-4o)
MemGraphRAG	Hier. (3-layer)	—	MS	Schema filter + adjud.	59.25%	Multi-hop QA
Entity-relation KG systems						
Mem0	KG (entity-relation)	—	GA	Dedup/merge	28.86 F1	LoCoMo (7B)
HippoRAG	KG (open)	—	GA	—	+3–10 F1	Multi-hop QA
CraniMem	KG (gated buffer)	—	GA	Replay consol.	−0.011 F1	HotpotQA noise
Temporal-aware systems						
RecallM	KG (temporal window)	✓	GA	Context revision	4× vs Vec	Temporal QA
MemoTime	KG (Tree of Time)	✓	GA	Temporal monoton.	—	Temporal reason.
TReMu	KG (neuro-symbolic)	✓	GA	Code-gen dates	—	Multi-session
RL & learned retrieval systems						
TGM	Procedural (3-layer)	—	RL	REINFORCE weights	0.426 EM	HotpotQA (4B)
Memory-R1	KG	—	RL	RL operations	—	Memory via RL
GNN & higher-order systems						
LP-RAG	Bipartite	—	GNN	—	+2.8% EM	Multi-hop QA
MemoGraph	KG + RGCN	—	GNN	—	86.9%	MATH (7B)
HyperMem	Hypergraph (3-level)	—	GA	—	—	—
Multi-agent & procedural systems						
Collabor Mem.	KG (shared)	—	AB	Access control	—	Multi-agent
PRAXIS	State-action tuples	—	GA	Real-time learn.	44.1%	REAL (web)
SkillRL	Hier. SkillBank	—	RL	Recursive evol.	+14%	ALFWorld
SimpleMem	Multi-view index	—	MS	Sem. compression	+26.4% F1	LoCoMo

bipartite chunk-query graph with mutual k-NN edges (retrieval via GNN link prediction outperforms embedding similarity by 1.6–2.8% EM), while AssoMem [20] anchors dialogue to automatically extracted clues, i.e. minimal overhead, but limited representational power.

At the other extreme, *multi-graph composites* maintain several orthogonal views simultaneously. MAGMA [21] connects event-nodes via four edge types (temporal, causal, semantic, entity) with dual-stream fast/slow consolidation. TM2RAG [22] extends this with five graphs unified through a meta graph.

Two emerging patterns push beyond pairwise relations. *Hypergraph memory* systems, such as HyperGraphRAG [23], HyperMem [24], HGMem [25], use hyperedges for n-ary facts and multi-step reasoning chains, though all treat the hypergraph as a data structure without higher-order message passing. *Procedural strategy graphs* encode agent trajectories directly: Trainable Graph Memory (TGM) [26] represents trajectories as a heterogeneous DAG optimized via REINFORCE, where the trained Qwen3-4B outperforms baseline Qwen3-8B, i.e. procedural graph memory substituting for model scale. PRAXIS [27] implements procedural memory for web agents. SkillRL [28] builds a hierarchical SkillBank.

Cutting across all patterns is the question of *temporal grounding*. Despite being the central promise of graph-based memory, 57% of surveyed papers implement no temporal mechanism and only 20% go beyond timestamps. Graphiti/Zep [10] provides the most complete solution via a bi-temporal model distinguishing event time from ingestion time, where every edge carries validity intervals. RecallM [29] uses temporal windows with Hebbian-inspired relation strength for a more effective belief updating than vector databases. MemoTime [30] organizes temporal reasoning into time trees, while TReMu [31], similarly to Graphiti/Zep, decouples mention time from event time. Biologically inspired mechanisms offer implicit temporal grounding. For example, Mnemosyne [32] applies exponential decay with spreading activation. MIRROR [54] validates fast encoding plus slow consolidation. MARTA [55] uses predictive entropy to arbitrate retrieval. However, a staleness gap persists, i.e. no system tracks when abstracted semantic knowledge becomes outdated.

3. From Graph Algorithms to Learned Retrieval

Memory retrieval over graphs spans a spectrum from training-free algorithms to fully learned neural approaches. The dominant approaches require no training. For example HippoRAG [9] implements Personalized PageRank as a neurobiologically inspired pattern-completion mechanism. GraphRAG [12] and MemFly use Leiden for community detection. GAM applies exponential recency decay. These algorithms work on any graph at 10–150 ms, explaining why every deployed system adopts them. Production systems go further by combining multiple pathways. E.g. Hindsight [11] executes semantic search, BM25, graph traversal, and temporal filtering simultaneously, fusing via cross-encoder. SimpleMem [33] achieves improvement over Mem0 while reducing tokens via three-stage semantic compression. Beyond heuristics, RL-based approaches learn to optimize different aspects of graph memory, such as edge weights [26], memory operations [34], graph construction [35], and unified retrieval-reasoning [36]. D-RAG [37] enables joint retriever-generator optimization through Gumbel-Softmax subgraph sampling. At the far end of the spectrum, only a tiny fraction of the surveyed systems apply GNN message passing, for example LP-RAG [19] recasts retrieval as inductive link prediction, MemoGraph [38] encodes evolving reasoning graphs. The scarcity reflects fundamental tensions, i.e. cold start, no retrieval labels, graph evolution outpacing retraining, and diminishing returns on factual tasks where heuristics already achieve high accuracy.

A striking pattern across independently reported results is that graph-structured memory disproportionately benefits smaller models, although we note that different benchmarks, base models, and evaluation protocols were used per study. GAM achieves +39% F1 improvement over Mem0 at 7B but only +7% at GPT-4o, and on LongDialQA this ratio reaches 17 times improvement. The trained Qwen3-4B with graph memory outperforms baseline Qwen3-8B showing that graph structure directly substitutes for model scale. GNN-RAG [39] shows a fine-tuned 7B model with graph retrieval matches GPT-4 on knowledge graph question answering tasks.

While controlled experiments across a unified benchmark are needed to confirm this trend, the reported results may be explained by four mechanisms: i) context window relief via focused subgraph retrieval; ii) explicit structure offloading reasoning that larger models derive implicitly; iii) token efficiency (GNN-RAG's reduction); and iv) strategic compensation from meta-cognitive priors in graph structure. Finally, graph memory must evolve as agents accumulate experience, i.e. consolidation merges episodic observations into compressed semantic representations, and SkillRL [28] achieves recursive skill co-evolution with significant token compression versus raw trajectory storage.

4. Geometric Representations for Agent Memory

The preceding sections reveal a consistent pattern: graph-structured memory improves retrieval and reasoning, yet three structural limitations persist, i.e. cognitive accuracy ceilings on multi-hop tasks, a temporal grounding deficit in the majority of systems, and semantic collapse as memory graphs grow. These limitations are not algorithmic but *geometric*: they arise from embedding inherently non-flat structures into flat space. We now turn to techniques from geometric deep learning that directly address each of these failure modes.

All graph-based memory systems surveyed above embed their graphs in flat Euclidean space. Yet recent work on graph neural networks, knowledge graph embeddings, and topological deep learning [56] shows that non-Euclidean representations, such as hyperbolic embeddings, discrete Ricci curvature, Lie group manifolds, can dramatically improve performance on hierarchical and temporal data. Since graph-structured memory is fundamentally a retrieval problem, i.e. finding the right subgraph at the right time, these techniques are directly applicable as foundational components of memory architectures.

Selective retrieval over a knowledge graph fails when the graph topology itself creates bottlenecks. *Discrete Ricci curvature* [57, 40] provides a per-edge diagnostic for this: edges within tight clusters have positive curvature, while edges bridging distant communities have negative curvature. Negatively curved edges cause *over-squashing* [58], i.e. information from exponentially many source nodes must squeeze through a single bridge edge, causing multi-hop retrieval to suffer exponential signal decay. This offers a structural explanation for the cognitive accuracy ceiling below 25%. Three concrete applications emerge: i) Forman-Ricci curvature provides a training-free, $O(|E|)$ diagnostic that flags edges where retrieval is likely to degrade; ii) Stochastic Discrete Ricci Flow (SDRF) [40] offers a principled geometric alternative to Leiden-based community detection for hierarchical consolidation; and iii) structural lifting from graphs to hypergraphs [59, 41] can promote bottleneck edges to hyperedges, restoring multi-hop connectivity.

Selective retrieval also fails when the embedding space itself cannot separate what the graph distinguishes. As memory grows, Euclidean spaces suffer from *semantic collapse*: hierarchical structures are compressed into a low-dimensional subspace because Euclidean volume grows polynomially while tree-like knowledge branching grows exponentially [60]. Gao and Metaxas [61] formalize this as semantic shift, i.e. as diversity increases, pooled representations drift away from every individual embedding. *Hyperbolic space*, where volume grows exponentially, provides a geometric remedy by naturally separating hierarchy levels: abstract topics embed near the origin, episodes in the middle ring, and specific facts near the boundary. HyperbolicRAG [42] fuses Euclidean and hyperbolic rankings for hierarchy-aware retrieval. HypRAG [43] develops fully hyperbolic transformers in the Lorentz model. HELM [44] shows LLM token embeddings already exhibit substantial negative Ricci curvature, suggesting that LLM representations are naturally hyperbolic and motivating matching the memory embedding space to the model's intrinsic geometry. HyperVC [62] applies variable-curvature hyperbolic spaces to temporal KG snapshots. However, HyperGCL [63] shows that naive contrastive learning in hyperbolic space still causes dimensional collapse, meaning collapse-aware objectives are required, such as HypRAG's Outward Einstein Midpoint pooling operator. Since real memory graphs exhibit topological heterogeneity, i.e. some regions are tree-like, others cyclical or clique-dense, a single curvature may not suffice. GraphMoRE [64] addresses this with a mixture of Riemannian experts that assigns spherical, hyperbolic, or Euclidean geometry per node.

Cross-session memory further requires temporal reasoning: retrieving not just what was stored but when, and how knowledge evolved. In temporal KGs, entities, relations, and timestamps live in fundamentally different spaces, making direct fusion difficult. *Lie groups* such as $SO(n)$ (rotation matrices) offer a solution: their tangent space, i.e. the Lie algebra, is an ordinary vector space, so projecting via the logarithmic map lets heterogeneous factors be added, scaled, and processed by standard neural network layers. Li et al. [46] apply this using $SO(2)$ Givens rotations to unify temporal KG factor embeddings. TeAST [45] maps temporal relations onto Archimedean spiral geometry. TempHypE [65] unites Poincaré ball embeddings with Neural ODEs for continuous-time evolution. RicciKGE [66] co-evolves Ricci flow with KG embeddings, where curvature smooths bottleneck regions while updated embeddings reshape curvature. At the foundation-model scale, RiemannGFM [67] learns cross-domain transferable Riemannian graph representations, suggesting that zero-shot geometric encoding of memory traces may become feasible without per-agent supervised training.

To make this concrete, consider how Graphiti/Zep, which is the strongest bi-temporal system in our survey, could be geometrically enhanced. Its entity-relation KG already carries validity intervals on every edge. First, computing Forman-Ricci curvature over this graph (an $O(|E|)$ operation requiring no training) would flag negatively curved bridge edges where multi-hop retrieval is likely to degrade, providing an automatic diagnostic for the cognitive accuracy ceiling. Second, replacing its Euclidean entity embeddings with Poincaré ball embeddings would let abstract topics (e.g., *project goals*) embed near the origin while specific episodic facts (e.g., *meeting notes from June 3*) sit near the boundary, preserving hierarchical separation as the memory graph grows. Third, its temporal validity intervals could be fused with entity and relation embeddings via $SO(2)$ Lie group rotations, enabling the system to answer not just *what* but *when* and *how beliefs evolved* without dedicated temporal query logic. None of these modifications require retraining the underlying LLM; they operate on the memory layer itself.

These geometric techniques are already proven in retrieval and KG reasoning. The open challenge is integrating them into persistent, cross-session memory architectures with evolving graphs.

5. Discussion

Graph-structured memory reframes the persistence problem: instead of compressing everything into the context window, store knowledge in a graph and retrieve selectively. Our survey shows this paradigm works, with multi-strategy retrieval reaching top performances on long memory evaluation benchmarks. Across independently reported results graph memory appears to disproportionately benefit smaller models, narrowing the gap to frontier performance at significantly lower cost.

Futher, the convergence of evidence from geometric deep learning suggests that the structural limitations identified in our survey, i.e. cognitive accuracy ceilings, semantic collapse, temporal reasoning deficits, are not merely algorithmic shortcomings but consequences of a *geometric mismatch* between the inherently non-flat topology of memory graphs and the Euclidean spaces in which they are embedded. This reframing shifts the research focus from better retrieval heuristics toward principled geometric foundations for memory architectures.

In terms of practical implications of our survey, we find that several geometric techniques require no model retraining and can be integrated into existing systems as diagnostic or retrieval layers. Forman-Ricci curvature provides a bottleneck diagnostic applicable to any graph-memory system. Hyperbolic embeddings offer a drop-in replacement for Euclidean entity representations that preserves hierarchical separation as memory grows. Mixed-curvature product spaces accommodate the topological heterogeneity of real memory graphs without requiring a single global curvature choice.

To this end, several challenges remain. Current systems vary widely in graph types, node/edge semantics, and temporal models. A unified ontology specifying how entities, events, temporal intervals, and higher-order groupings compose would enable interoperability and memory transplantation across architectures [47]. Graph algorithms dominate practice while GNN-based message passing remains limited to a handful of systems. However, whether learned structural representations can break the cognitive ceiling is an open question. Collaborative memory is largely unexplored, yet the same relational structure that enables rich retrieval also enables adversarial injection [50, 48, 49]. Memory quality deserves attention beyond downstream accuracy: diagnostic analysis [51] reveals that retrieval quality, not write-time sophistication, remains the primary bottleneck, and filtering low-frequency schema triples actually improves accuracy. Scalability is a practical concern, with systems spanning wide latency ranges; at lifetime scale, approximate retrieval, compression, and asynchronous processing [52] become critical. Finally, geometric retrieval techniques, i.e. curvature diagnostics, hyperbolic embeddings, mixed-curvature manifolds, and continuous-time dynamics, are proven in retrieval and knowledge graph reasoning but await systematic integration into persistent, cross-session memory architectures.

Finally, we note that this work is a survey and conceptual synthesis, i.e. the proposed geometric extensions have not been empirically validated within agent memory systems. Our model-size equalizer analysis aggregates results across independently published benchmarks with different base models, datasets, and evaluation protocols, and controlled experiments are needed to confirm the observed trends. The survey scope is limited to 2023–2026, which captures the current wave of graph-memory research but may miss relevant earlier work on persistent knowledge representations. Despite these limitations, the consistent alignment between the structural deficits identified in our survey and the capabilities demonstrated by geometric techniques in adjacent fields provides, in our view, a well-motivated foundation for the proposed research agenda.

6. Conclusion

We surveyed graph-based memory architectures for LLM agents, identifying six architectural patterns, a retrieval spectrum from training-free graph algorithms to GNN message passing, and a persistent temporal grounding deficit across the field. Building on these findings, we argued that non-Euclidean geometric techniques, including discrete Ricci curvature, hyperbolic embeddings, Lie group manifolds, and mixed-curvature product spaces, are directly applicable to the structural limitations we identified, and illustrated this with a concrete enhancement scenario for bi-temporal memory systems. Integrating these geometric foundations into persistent, cross-session memory architectures is the defining open challenge, and addressing it would move the field toward truly unified agent memory capable of the structured recall that cognitive continuity demands.

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